AlphaProteo Reading Summary

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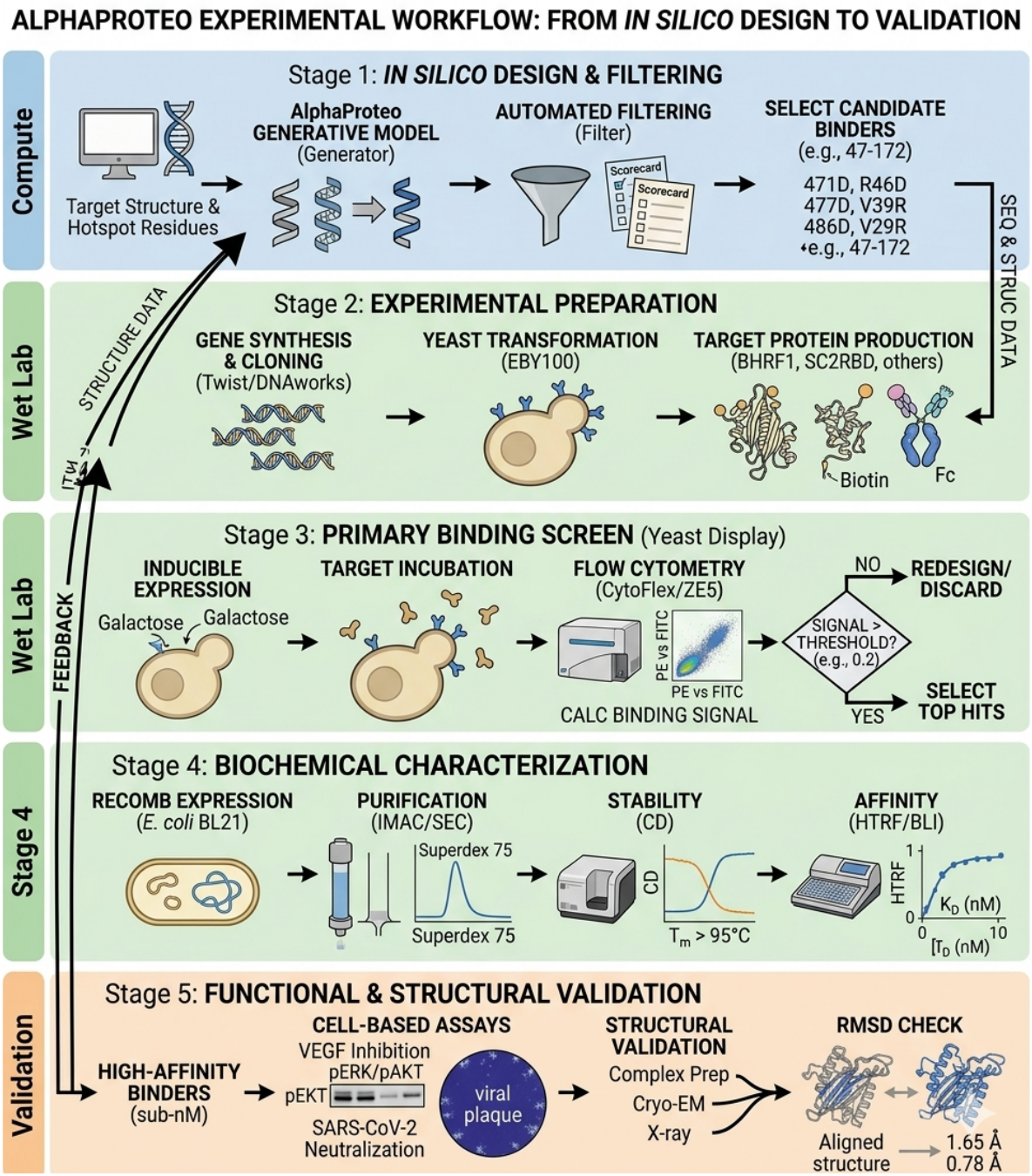

Figure 1: Workflow(generated by Gemini)

1 Introduction

Because AlphaProteo is not open source until now, I may talk more about the experimentation part of this paper (or report) on Arxiv `/https://arxiv.org/abs/2409.08022`.

In these paper, they choose 8 kinds of specific protein to design their binder using AlphaProteo. After designing, they need to validate the whether to combine, the binding affinity, the conformation and the thermal stability.

For binder purification, I plan to skip because there aren't too many physical principles involved.

BHRF1, SC2RBD, IL-7RA, PD-L1, TrkA, IL-17A, VEGF-A, TNFα

2 Target protein expression and purification

For IL-7RA(21-239), TrkA(34-423), PD-L1(19-239), VEGF-A(27-191), IL-17A(24-155) and VEGF165, they purchased them from other company.

2.1 Non-Equilibrium Nature

Protein expression is not an equilibrium process: it is continuously driven by energy (NTP/ATP) flux. A simple kinetic description is:

$$\frac{d[P]}{dt} = k_{\text{prod}} - k_{\text{deg}}[P]$$

Here, k_{prod} depends on transcription and translation rates which can be controlled by IPTG, while k_{deg} reflects cellular degradation.

Intuition: The cell maintains a steady-state protein concentration, not a thermodynamic equilibrium. Energy input keeps the system dynamically active.

2.2 IPTG Modulates Transcription Probability

IPTG works by binding LacI, changing its conformation, and reducing its affinity for DNA:

$$P_{\text{DNA unbound}} = \frac{1}{1 + e^{-\Delta G/kT}}$$

The transcription rate scales with the probability that the promoter is free:

$$k_{\text{transcription}} \propto P_{\text{DNA unbound}}$$

Intuition: IPTG does not just "switch on" transcription; it tunes the promoter occupancy continuously, effectively setting k_{prod}.

2.3 Temperature and Folding Kinetics

After translation, proteins can fold correctly or aggregate:

$$\text{nascent protein} \begin{cases} \xrightarrow{k_{\text{fold}}} \text{native} \\ \xrightarrow{k_{\text{agg}}} \text{aggregate} \end{cases}$$

Temperature affects both translation and folding:

- High T: faster translation $\rightarrow$ less time for proper folding $\rightarrow$ misfolding increases
- High T: conformational fluctuations $\uparrow \rightarrow$ aggregation increases
- Low T: slower translation $\rightarrow$ more time for co-translational folding $\rightarrow$ aggregation decreases

Functional protein yield:

$$[P_{\text{functional}}] \propto k_{\text{prod}} \cdot \frac{k_{\text{fold}}}{k_{\text{fold}} + k_{\text{agg}}}$$

Intuition: Temperature shifts the balance of flux through folding vs aggregation pathways, not just "slows things down".

2.4 Unified Kinetic Picture

A complete view combines production, folding, and aggregation:

$$\begin{cases} \frac{d[P_{\text{nascent}}]}{dt} = k_{\text{prod}}(\text{IPTG}, T) - (k_{\text{fold}} + k_{\text{agg}})[P_{\text{nascent}}] \\ \frac{d[P_{\text{functional}}]}{dt} = k_{\text{fold}}[P_{\text{nascent}}] - k_{\text{deg}}[P_{\text{functional}}] \\ \frac{d[P_{\text{aggregate}}]}{dt} = k_{\text{agg}}[P_{\text{nascent}}] \end{cases}$$

Intuition: The final functional protein concentration emerges from a competition between production, folding, and irreversible aggregation, modulated by IPTG and temperature.

3 Binding Detection as a Signal Transduction System

3.1 From Molecular Binding to Observable Signal

The goal of the assay is to convert an invisible molecular binding event into a measurable fluorescence signal. This is achieved through a cascade:

$$\text{binder} + \text{target} \rightleftharpoons \text{complex}$$

$$\text{complex} \xrightarrow{\text{biotin-streptavidin}} \text{fluorescent signal (PE)}$$

Intuition: Binding itself is not directly observable. The system introduces a molecular "linker" (biotin) and a fluorescent reporter (PE) to transform binding into light.

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3.2 Fluorescence as a Proxy for Physical Quantities

Two fluorescence channels are measured:

$$F_{\text{FITC}} \propto [\text{binder expression}]$$

$$F_{\text{PE}} \propto [\text{binder-target complex}] + F_{\text{background}}$$

Intuition: FITC measures how much binder is present, while PE measures how much target is captured. The experiment separates expression from binding.

3.3 Binding Equilibrium and Signal Generation

Binding between binder (B) and target (T) can be described as:

$$B + T[k_{\text{off}}]k_{\text{on}}BT$$

At quasi-steady state:

$$[BT] \propto \frac{[B][T]}{K_D}$$

The measured fluorescence becomes:

$$F_{\text{PE}} \sim [BT] + F_{\text{non-specific}}$$

Intuition: Strong binders (low K_D) produce more complexes and therefore stronger fluorescence. However, background signals are always present and must be corrected.

3.4 Background Subtraction and Signal Isolation

To isolate true binding, four populations are considered:

$$\text{signal} = (\log_{10} F_{\text{FITC+,+target}} - \log_{10} F_{\text{FITC-,+target}}) - (\log_{10} F_{\text{FITC+,-target}} - \log_{10} F_{\text{FITC-,-target}})$$

Intuition: The first term captures total binding-related signal, while the second term removes contributions from non-specific interactions and expression-dependent artifacts. The result is a net binding signal.

3.5 Signal Decomposition

The observed fluorescence can be decomposed as:

$$F_{\text{PE}} = F_{\text{specific}} + F_{\text{non-specific}} + F_{\text{expression-dependent}}$$

Thus:

$$\text{binding signal} \approx F_{\text{specific}}$$

Intuition: The analysis pipeline removes all contributions except the fluorescence arising directly from binder-target interaction.

3.6 Unified Detection Model

Combining expression, binding, and detection:

$$\begin{cases} [B] \sim \text{expression level (FITC)} \\ [BT] \sim \frac{[B][T]}{K_D} \\ F_{\text{PE}} \sim [BT] + \text{background} \\ \text{signal} \sim \log([BT]) - \text{background correction} \end{cases}$$

Intuition: The final measured signal is not a direct readout of binding, but the result of a multi-step transformation from molecular interaction to fluorescence, followed by statistical correction.

4 Interface Validation, Competition, and Specificity Assays

4.1 Interface Mutation as a Perturbation Test

To validate whether the predicted binding interface is correct, mutations are introduced at the binder–target interface:

$$\text{binder}_{\text{WT}} \rightarrow \text{binder}_{\text{mut}}$$

Specifically, hydrophobic residues at the interface are replaced with charged residues:

$$\text{hydrophobic (A, V, L, I)} \;\rightarrow\; \text{charged (D, E, R, K)}$$

Binding is then re-measured using the same fluorescence-based assay:

$$F_{\text{PE}}^{\text{mut}} \propto [\text{binder}_{\text{mut}} \cdot \text{target}]$$

Experiment: Mutant binders are expressed on yeast and subjected to the same binding screen as the wild-type designs.

Intuition: Hydrophobic residues often stabilize protein–protein interfaces. Replacing them with charged residues disrupts packing and introduces electrostatic penalties. If binding decreases significantly, it indicates that the mutated residues are part of the true interface.

4.2 Competition Assay for Binding Site Overlap

To determine whether the binder targets a specific site, a competitor protein with known binding to the same target is introduced:

$$\text{binder} + \text{target} + \text{competitor}$$

Binding becomes a competitive process:

$$\text{target} + \text{binder} \rightleftharpoons BT$$

$$\text{target} + \text{competitor} \rightleftharpoons CT$$

The observed binding signal:

$$F_{\text{PE}} \propto [BT]$$

Experiment: Yeast-displayed binders are incubated with biotinylated target in the presence or absence of a competitor protein. Binding is measured via streptavidin-PE fluorescence as before.

Intuition: If the binder and competitor share the same binding site, the competitor reduces the available free target, decreasing $[BT]$ and thus the PE signal. If the signal remains unchanged, the two likely bind distinct sites.

4.3 Avidity-Enhanced Detection for Weak Binding

For weak interactions, multivalent complexes are formed prior to binding:

$$\text{target} + \text{detection reagents} \rightarrow \text{multivalent complex}$$

Experiment: Target proteins are pre-incubated with fluorescent detection components before being added to cells.

Intuition: Multivalent interactions increase effective binding strength (avidity), stabilizing otherwise transient interactions and enhancing detectable signal.

4.4 Specificity Measurement via HTRF

Binding specificity is assessed using a homogeneous time-resolved fluorescence (HTRF) assay:

$$F_{\text{HTRF}} \propto [\text{binder} \cdot \text{target}]$$

The assay is performed at fixed concentrations:

$$[\text{binder}] = 1\,\text{nM}, \quad [\text{target}] = 100\,\text{nM}$$

Experiment: Each binder is tested against its intended target (and potentially other proteins) using HTRF, which measures binding via energy transfer between fluorophores.

Intuition: By using low binder concentration and excess target, the assay emphasizes selective binding. A highly specific binder will produce strong signal only with its intended target, while non-specific binders will show broader interactions.

4.5 Integrated Interpretation

The three assays probe complementary aspects of binding:

$$\begin{cases} \text{Mutation} \rightarrow \text{interface validity} \\ \text{Competition} \rightarrow \text{binding site overlap} \\ \text{Specificity} \rightarrow \text{target selectivity} \end{cases}$$

Intuition: Together, these experiments ensure that binding is structurally correct (interface), spatially meaningful (site), and biologically relevant (specificity), rather than arising from non-specific interactions.

5 Why the binding model is divided by B (HTRF signal normalization)

5.1 Exact 1:1 binding equilibrium

For a reversible bimolecular interaction:

$$A + B \rightleftharpoons AB$$

the equilibrium is governed by:

$$K_D = \frac{[A][B]}{[AB]}$$

with mass conservation constraints:

$$A_0 = A + AB, \quad B_0 = B + AB$$

Solving exactly for the equilibrium complex concentration yields:

$$[AB] = \frac{(A_0 + B_0 + K_D) - \sqrt{(A_0 + B_0 + K_D)^2 - 4A_0B_0}}{2}$$

Intuition: The system is fully constrained by thermodynamics and mass conservation. No assumption of excess ligand is made, so ligand depletion is explicitly accounted for.

5.2 HTRF signal as a proxy for complex occupancy

In HTRF measurements, the observed signal is proportional to the number of formed complexes:

$$R \propto [AB]$$

However, instead of working with absolute complex concentration, the signal is normalized to the total binder concentration:

$$R = R_{\max} \cdot \frac{[AB]}{B_0}$$

Intuition: The readout is interpreted as fractional occupancy of the binder population rather than absolute binding quantity.

5.3 Final HTRF model equation

Substituting the exact solution for $[AB]$ into the occupancy definition $\theta = [AB]/B_0$ gives:

$$R = \frac{R_{\max}}{2B_0}\left(A_0 + B_0 + K_D - \sqrt{(A_0 + B_0 + K_D)^2 - 4A_0B_0}\right)$$

5.4 Intuition

The HTRF signal reflects equilibrium occupancy rather than absolute fluorescence intensity:

$$R \sim P(\text{binder is bound to target})$$

Dividing by B_0 converts the physical quantity from absolute complex concentration into a normalized occupancy variable, enabling comparison across experiments and allowing accurate extraction of K_D even when the binding partner is not in large excess.

6 Kinetic Binding Affinity Measurement by BLI

Bio-layer interferometry (BLI) measures protein–protein interactions by tracking real-time changes in optical interference caused by mass accumulation on a biosensor surface.

6.1 Experimental Design

A fixed concentration of ligand (8His-tagged binder) is immobilized onto Ni-NTA biosensors:

$$\text{Ligand (B)} \rightarrow \text{sensor surface}$$

Target proteins (analyte, A) are prepared in a multi-point dilution series and introduced into solution.

The experiment consists of three sequential phases:

- **Loading:** immobilization of ligand on sensor surface
- **Association:** binding of analyte to immobilized ligand
- **Dissociation:** removal of free analyte to observe complex decay

Intuition: The sensor surface acts as a quasi-2D reaction platform where binding and unbinding events are continuously monitored in real time.

6.2 Kinetic Model

Binding is described by a simple reversible reaction:

$$A + B \rightleftharpoons AB$$

with kinetic rates:

$$\frac{d[AB]}{dt} = k_{\text{on}}[A][B] - k_{\text{off}}[AB]$$

During dissociation (no free analyte):

$$\frac{d[AB]}{dt} = -k_{\text{off}}[AB]$$

Intuition: Association reflects competition between binding and unbinding, while dissociation isolates the intrinsic stability of the complex.

6.3 Affinity Extraction

The equilibrium dissociation constant is derived from kinetic parameters:

$$K_D = \frac{k_{\text{off}}}{k_{\text{on}}}$$

Intuition: Affinity is not directly measured, but emerges from the ratio of microscopic association and dissociation rates governing the dynamic equilibrium.

6.4 Integrated Interpretation

BLI does not measure affinity directly, but reconstructs it from time-resolved binding trajectories:

$$R(t) \Rightarrow k_{\text{on}}, k_{\text{off}} \Rightarrow K_D$$

Intuition: The system converts dynamic surface binding behavior into kinetic parameters, which together define equilibrium binding strength.

7 First-principles interpretation of Circular Dichroism (CD) spectroscopy

7.1 CD as a chiral optical response function

Circular dichroism measures the differential absorption of left- and right-circularly polarized light:

$$\Delta\varepsilon(\lambda) = \varepsilon_L(\lambda) - \varepsilon_R(\lambda)$$

From a first-principles perspective, this quantity is not a structural descriptor, but a response function of a chiral electronic system:

$$CD(\lambda) \sim \left\langle \Psi \middle| \hat{O}_{CD}(\lambda) \middle| \Psi \right\rangle$$

where $\hat{O}_{CD}$ is the chiral optical transition operator, and $|\Psi\rangle$ is the electronic state of the peptide backbone embedded in a solvent dielectric field.

7.2 Protein as a coupled oscillator network

Each peptide bond contributes an electronic transition dipole moment:

$$\mu_i(t)$$

These dipoles are not independent, but are coupled through excitonic interactions:

$$H = \sum_i \epsilon_i + \sum_{i \neq j} J_{ij}$$

where J_{ij} describes dipole–dipole coupling between transition oscillators.

The CD signal arises from interference between coupled transitions:

$$CD \sim \sum_{i,j} \langle \mu_i \cdot \mu_j \rangle$$

Thus, far-UV CD (260–190 nm) probes the collective excitonic structure of the peptide backbone rather than local secondary structure directly.

7.3 Thermal unfolding as a Boltzmann-weighted ensemble projection

At finite temperature, the protein samples a distribution of conformational states x:

$$P(x, T) = \frac{e^{-E(x)/kT}}{Z}$$

The observed CD signal at temperature T is therefore an ensemble average:

$$S(T) = \int P(x, T)\, O_{CD}(x)\, dx$$

where $O_{CD}(x)$ is the CD response of a given conformation.

Thus, thermal unfolding curves at 222 nm correspond to a projection of the folding free-energy landscape onto a single optical observable:

$$\theta_{222}(T) = \langle O_{CD}^{222} \rangle_T$$

7.4 Far-UV spectra as projection of excitonic manifold

The full wavelength-dependent CD spectrum is a projection of the excitonic Hamiltonian:

$$CD(\lambda) = \sum_n f_n\, \delta(\lambda - \lambda_n)$$

where f_n depends on transition dipole coupling geometry.

7.5 Baseline subtraction as removal of environmental response

The measured signal contains contributions from multiple sources:

$$S_{\text{measured}} = S_{\text{protein}} + S_{\text{buffer}} + S_{\text{instrument}}$$

Baseline subtraction defines the reference state:

$$S_{\text{protein}} = S_{\text{measured}} - S_{\text{buffer}}$$

This operation removes solvent and cuvette-induced optical background, isolating the intrinsic molecular response.

7.6 Thermal hysteresis as free-energy landscape test

Cooling and refolding experiments probe the reversibility of the conformational ensemble:

$$P_{\text{heat}}(x, T) \stackrel{?}{=} P_{\text{cool}}(x, T)$$

Deviations indicate:

- multiple metastable basins (kinetic trapping),
- aggregation (irreversible landscape deformation),
- or hysteresis in folding free energy.

7.7 Conversion from instrument units to physical response

The raw CD signal recorded by the spectrometer is the optical rotation angle of circularly polarized light after passing through the sample, reported in millidegrees (mdeg):

$$\theta_{\text{obs}} \text{ (mdeg)}$$

This quantity is instrument-dependent, as it scales with path length, concentration, and instrument geometry, and therefore cannot be directly compared across experiments.

To obtain an intrinsic molecular property, the observed rotation is converted into the molar differential extinction coefficient, $\Delta\varepsilon$, which represents the differential absorption of left- and right-circularly polarized light per mole of chromophore:

$$\Delta\varepsilon = \frac{\theta_{\text{obs}}}{c \cdot l \cdot C}$$

where c is the concentration, l is the optical path length, and C is a unit conversion constant.

In this form, the signal becomes independent of experimental setup and reflects an intrinsic property of the molecular electronic structure.

Thus, the transformation:

$$\theta_{\text{obs}} \;\rightarrow\; \Delta\varepsilon \text{ (M}^{-1}\text{cm}^{-1}\text{)}$$

corresponds to converting an instrument readout (rotation angle) into a material response function that can be directly compared across proteins, conditions, and experiments.

7.8 Unified interpretation

Overall, CD spectroscopy measures a temperature-dependent projection of a protein's conformational ensemble onto a chiral excitonic response operator:

$$CD(T, \lambda) = \left\langle \hat{O}_{CD}(\lambda) \right\rangle_{P(x,T)}$$

where temperature modulates the underlying Boltzmann distribution, and the spectrometer measures its optical projection rather than structure directly.